

UNITED STATES PATENT APPLICATION

APPLICANT / ASSIGNEE: B-Lawz Music LLC

COUNTRY OF RESIDENCE: United States of America

ATTORNEY DOCKET NO.: BLM-007

TITLE OF INVENTION

TECHNICAL FIELD

The present invention relates to calibration systems for artificial intelligence content quality scoring, and more particularly to a calibrator that continuously bridges a large-scale multi-domain training dataset corpus with a local content quality scoring pipeline by querying the training infrastructure for current model capability signals, merging content generation signals with training depth weights, updating calibrated scoring dimension weights and acceptance thresholds, and maintaining the LLM serving infrastructure in a warm state to minimize first-inference latency.

BACKGROUND OF THE INVENTION

Content quality scoring systems used in autonomous publication pipelines typically employ static scoring weight vectors – fixed coefficients assigned to dimensions such as engagement potential, hook strength, sentiment, and clarity. These static weights do not adapt to the current state of the underlying AI model: a model with 10 hours of training experience may produce content where engagement potential is the most reliable predictor of quality, while a model with 10,000 simulated years of experience may have learned to optimize all dimensions simultaneously and the relative weighting should shift accordingly.

A further challenge is that static thresholds do not reflect the absolute capability ceiling of the current trained model. A threshold of 81/100 is appropriate when the model can realistically achieve that score; if the current model can only achieve 70 as its best output, an 81 threshold will perpetually reject all content regardless of how many retry rounds are attempted. Conversely, a model that routinely achieves 88 or higher would benefit from a raised threshold that pushes it toward its full capability.

Existing content scoring calibration approaches use offline training runs to fit scoring weights, require significant human annotation effort, and do not update in response to live model training events. They also do not account for the quality of the training infrastructure itself: whether base model weights are synced, whether calibration has run recently, and what the current training depth level implies about model capability.

There exists a need for a calibration system that (1) queries the training infrastructure for live model state signals, (2) merges content generation signals with model training depth weights, (3) updates calibrated scoring weights and acceptance thresholds at a configurable refresh interval, (4) exposes a training readiness profile that contextualizes quality outcomes, and (5) maintains the LLM inference infrastructure in a warm state to minimize cold-start latency.

SUMMARY OF THE INVENTION

The present invention provides a training-dataset-bridged content scoring calibrator comprising a sequential content signal query module, a parallel model state query module, a signal merging engine, a calibrated weight and threshold output layer, a training readiness classifier, and an LLM warmth maintenance subsystem.

In a first aspect, the invention provides a calibration method comprising: sequentially querying a training infrastructure for content quality signals with configurable inter-query delays to prevent LLM queue saturation; querying model state endpoints in parallel; merging generation signals and training depth weights into calibrated scoring dimension weights; and updating acceptance threshold and floor values based on the merged signals.

In a second aspect, the invention provides a readiness profiler that classifies the training infrastructure into four readiness

levels – cold, warming, ready, and optimal – based on the number of synced base model weights and calibration recency, and exposes a human-readable readiness summary for display in quality gate failure responses.

In a third aspect, the invention provides an LLM warmth subsystem that fires keepalive requests to the LLM serving infrastructure at a fixed interval to prevent cold-start states and ensure first-inference latency remains within a configurable bound.

DETAILED DESCRIPTION OF PREFERRED EMBODIMENTS

I. Training Infrastructure Architecture

The training infrastructure comprises an 8TB (and growing) multi-domain dataset corpus covering music industry content, social media management, and advertising performance. It exposes content signal endpoints (each involving LLM inference) and model state endpoints (GET endpoints, no LLM inference).

The calibrator interacts with this infrastructure through two query categories:

Content signal queries: 5 queries, sequential with 200ms inter-query gaps

These query the LLM for content quality signals across the five primary model domains.

Sequential execution prevents queue saturation on the single-threaded LLM server.

Model state queries: 4 queries, parallel via Promise.allSettled

These query lightweight GET endpoints for model training depth, weight sync status, and calibration history without invoking LLM inference.

II. Default Scoring Weight Vector

Ten scoring dimensions are defined with default weights:

engagement: 0.25 (predicted engagement rate – highest weight)
hookStrength: 0.18 (first-sentence attention capture)
callToActionEffectiveness: 0.13 (actionability of the post's ask)
sentiment: 0.10 (emotional resonance and positivity)
clarity: 0.08 (readability and message clarity)
brandAlignment: 0.08 (consistency with brand voice profile)
algorithmAlignment: 0.08 (platform algorithm optimization signals)
specificity: 0.05 (concreteness of claims and data)
emotionalArc: 0.03 (narrative arc from hook to CTA)
narrativeAuthenticity: 0.02 (domain-specific vocabulary authenticity)

These weights sum to 1.00 and are normalized after calibration adjustment to maintain this property.

III. Default Threshold Values

DEFAULT_THRESHOLD = 81 (90% of Veo's ~90 baseline score on this rubric)

DEFAULT_FLOOR = 73 (absolute minimum – approximately 90% of the gate threshold)

IV. Signal Merging Engine

The signal merging engine receives content generation signals from the sequential LLM queries and model training depth weights from the parallel model state queries. It produces calibrated weight adjustments as follows:

For each scoring dimension d :

```
base_weight(d) = DEFAULT_WEIGHTS[d]
generation_signal(d) = content signal query result for domain d
(normalized 0-1)
training_depth_weight(d) = model state depth signal for domain d
(normalized 0-1)
calibrated_weight(d) = base_weight(d) × (1 + α ×
generation_signal(d)) × (1 + β × training_depth_weight(d))
```

where α and β are calibration sensitivity constants. After all dimensions are adjusted, weights are renormalized to sum to 1.00.

Calibrated threshold values are computed similarly:

```
if training_depth is high and generation signals are strong:
    threshold may rise above 81
if training infrastructure is in warming state: threshold
    remains at defaults
```

V. Calibration Refresh Cycle

Calibration runs at startup and every 6 hours thereafter (CALIBRATION_TTL_MS = 6 × 3600 × 1000 ms). This interval is chosen to:

- Be short enough to capture new model training runs that complete within a session cycle
- Be long enough to avoid excessive load on the training infrastructure LLM server

A mutual exclusion guard prevents concurrent calibration runs. Calibration is triggered non-blockingly when the quality gate encounters a cold or warming readiness state.

VI. LLM Warmth Subsystem

LLM inference servers typically enter a cold state after a period of inactivity, causing the first inference request to experience a latency spike (commonly 30-60 seconds for large

models). The warmth subsystem fires a lightweight keepalive request to the LLM serving endpoint every 90 seconds:

```
startMaxCoreLLMWarmth():
```

Sets an interval at 90 seconds.

Each interval fires a minimal content signal request to the LLM.

The response is discarded; only the warmth effect on the server is used.

First inference attempts after warmth is active return in approximately 6 seconds.

The warmth subsystem is started at calibrator initialization and runs for the lifetime of the process.

VII. Training Readiness Profiler

The readiness profiler examines four named base model weights (social_base, advertising_base, content_base, engagement_base) and the calibration status flag:

```
cold: 0 weights synced AND calibration not run
```

Summary: "No MaxCore base weights synced yet and calibration has not run – system is starting up"

```
warming: Some weights synced OR calibration pending
```

Summary: "{n}/4 MaxCore base models present, calibration {complete/pending} – scores will

improve as the training simulator and memory sync complete their first cycle"

```
ready: All 4 weights synced, calibrated at defaults
```

Summary: "All 4 MaxCore base models synced, calibrated at defaults (gate=81, floor=73) –

scores will rise further as training simulator accumulates sessions"

optimal: All 4 weights synced, calibrated ABOVE defaults
(gate > 81)

Summary: "All 4 MaxCore base models synced and
training-calibrated (gate={g}, floor={f}) –
maximum quality capability active"

The readiness profile is cached for 60 seconds to prevent
re-querying on every quality gate invocation. It is returned
with every quality gate failure response, providing contextual
explanations and remediation guidance.

VIII. Per-User Engagement Feedback Integration

The calibrator's threshold output feeds into the quality gate's
per-user adaptive threshold system (see Patent No. 4). The
quality gate applies the calibrated threshold as the floor for
user-specific threshold adaptation: the per-user threshold may
rise above the calibrated gate threshold via engagement
feedback, but may never fall below it.

When calibration raises the gate threshold above the static
default (indicating the model has improved), all per-user
thresholds that are currently at the old default are silently
raised to the new calibrated default, ensuring that improvement
in model capability is immediately reflected in the quality gate
behavior for all users.

IX. Public API

The calibrator exposes four public functions:

getCalibratedWeights() → ScoreWeights

Returns the current calibrated weight vector, defaulting to
DEFAULT_WEIGHTS if uncalibrated.

getCalibratedThresholds() → CalibratedThresholds

Returns the current calibrated gate and floor thresholds,
defaulting to {81, 73}.

`isCalibrated()` → boolean

Returns true if calibration has run at least once in the current process lifetime.

`runCalibration()` → Promise

Triggers an immediate non-blocking calibration cycle if one is not already in progress

and the TTL has elapsed.

These functions are consumed by the content quality gate to determine whether to use calibrated or default thresholds and weights for each quality gate invocation.

CLAIMS

1. A computer-implemented calibration system for a content quality scoring pipeline, the system comprising: a processor; and a non-transitory computer-readable medium storing instructions that, when executed by the processor, cause the processor to: sequentially query a training infrastructure for a plurality of content quality signals with a configurable inter-query delay between each query, preventing saturation of a single-threaded language model inference server; concurrently query a plurality of model state endpoints that do not require language model inference; merge the content quality signals and model state depth weights into a calibrated scoring weight vector for a plurality of content scoring dimensions; and update a calibrated acceptance threshold and a calibrated pressure floor based on the merged signals, wherein the calibrated threshold rises above a static default when training depth signals indicate model capability above the default threshold's baseline.
2. The system of claim 1, wherein the configurable inter-query delay between sequential content signal queries is 200 milliseconds, preventing simultaneous requests to a single-threaded language model server that would otherwise queue multiple inference requests and cause elevated response latency.
3. The system of claim 1, wherein the calibration cycle runs at startup and repeats at a configurable refresh interval, wherein the refresh interval is chosen to be short enough to capture new model training runs within a session cycle and long enough to avoid excessive load on the training infrastructure.

4. The system of claim 1, further comprising a mutual exclusion guard that prevents concurrent calibration cycles from executing simultaneously, ensuring that each calibration cycle completes before the next is initiated.
5. The system of claim 1, further comprising a training readiness profiler that classifies the training infrastructure into one of a plurality of readiness levels based on the number of base model weight sets present in the infrastructure and the recency of the last calibration run.
6. The system of claim 5, wherein the plurality of readiness levels comprises a cold level indicating no base weights synced and calibration not run; a warming level indicating partial weight sync or calibration pending; a ready level indicating full weight sync with default thresholds; and an optimal level indicating full weight sync with training-calibrated thresholds exceeding the static defaults.
7. The system of claim 5, wherein the readiness profile is cached in process memory for a configurable duration to prevent re-querying on every quality gate invocation.
8. The system of claim 1, further comprising an LLM warmth maintenance subsystem that fires lightweight keepalive requests to the language model inference endpoint at a fixed interval during the lifetime of the process, preventing cold-start latency spikes on inference requests that follow periods of inactivity.
9. The system of claim 8, wherein the fixed keepalive interval is 90 seconds, and wherein the keepalive request result is discarded with only the server-side inference warmth effect retained.

10. The system of claim 1, wherein the content scoring dimensions comprise at least ten dimensions including engagement potential, hook strength, call-to-action effectiveness, sentiment, clarity, brand alignment, algorithm alignment, specificity, emotional arc, and narrative authenticity, each assigned a weight in a calibrated weight vector that sums to one.
11. The system of claim 10, wherein calibrating the weight vector comprises computing a base weight adjustment for each dimension from a content signal query result and a training depth weight, multiplying the base weight by a function of both factors, and renormalizing all dimension weights to sum to one after adjustment.
12. The system of claim 1, further comprising a non-blocking calibration trigger that initiates a calibration cycle when the quality gate encounters a cold or warming readiness state, enabling the next quality gate invocation to benefit from calibrated thresholds without delaying the current invocation.
13. A method of calibrating a content quality scoring system using live training infrastructure signals, the method comprising: at startup and at periodic refresh intervals, querying a training infrastructure for content generation quality signals using sequential requests with inter-query delays; querying training depth and model state signals concurrently; computing calibrated scoring dimension weights by adjusting default weights using functions of the content quality and training depth signals; computing calibrated acceptance threshold and floor values based on the training depth and model state signals; caching the calibrated weights and thresholds in process memory for use by a downstream content quality gate; and providing the calibrated weights and thresholds as defaults to a per-user adaptive threshold

system that can raise individual user thresholds above the calibrated default but not below it.

14. The method of claim 13, further comprising classifying the training infrastructure readiness into a plurality of levels and including a human-readable readiness summary and remediation hint with each quality gate failure response.
15. The method of claim 13, further comprising raising per-user quality thresholds that are currently at the previous calibrated default to the new calibrated default when a calibration cycle raises the calibrated gate threshold, ensuring model capability improvements are immediately reflected in all user-specific quality gates.
16. The method of claim 13, further comprising maintaining a language model inference server in a warm state by firing keepalive requests at a fixed interval, reducing first-inference latency from a cold-start latency of tens of seconds to a warm-inference latency within a configurable bound.
17. A non-transitory computer-readable medium storing instructions that, when executed by a processor, implement: a sequential content signal querier configured to query a language model inference server for a plurality of content quality signals with a configurable inter-query delay; a parallel model state querier configured to concurrently query a plurality of model state endpoints that do not involve language model inference; a signal merger configured to compute calibrated scoring dimension weights as a function of content quality signals and model training depth weights, normalized to sum to one; a threshold calibrator configured to output a calibrated acceptance threshold and pressure floor based on merged signals; a readiness classifier

configured to determine a readiness level from the number of available base model weights and calibration recency; and a warmth manager configured to fire periodic keepalive requests to the language model inference endpoint to prevent cold-start latency.

18. The computer-readable medium of claim 17, wherein the sequential content signal querier fires queries to a plurality of model domain endpoints representing distinct functional domains of the training infrastructure, and wherein the inter-query delay prevents simultaneous queuing of multiple inference requests on the server.

19. The computer-readable medium of claim 17, wherein the threshold calibrator outputs an optimal readiness level when the calibrated gate threshold exceeds the static default threshold, indicating that model training has raised the quality capability above the baseline reference level.

20. The computer-readable medium of claim 17, wherein the warmth manager operates independently of calibration cycles, running continuously at its fixed keepalive interval regardless of whether a calibration cycle is in progress, and wherein keepalive requests are intentionally minimal in content to minimize server-side processing overhead while achieving the warmth maintenance objective.

ABSTRACT

A training-dataset-bridged content scoring calibrator continuously updates a content quality scoring pipeline by sequentially querying a training infrastructure for content quality signals (with 200ms inter-query delays to prevent LLM queue saturation), concurrently querying model state endpoints, and merging the results into calibrated scoring dimension weights and acceptance thresholds. Default weights across ten scoring dimensions (engagement, hook strength, call-to-action effectiveness, sentiment, clarity, brand alignment, algorithm alignment, specificity, emotional arc, and narrative authenticity) are adjusted by functions of content generation quality signals and model training depth weights, then renormalized to sum to one. Calibrated acceptance thresholds rise above the static 81/100 Veo-score-derived default when training depth signals indicate model capability above baseline. A training readiness profiler classifies the infrastructure into four levels (cold, warming, ready, optimal) and provides human-readable remediation hints with quality gate failures. An LLM warmth subsystem fires keepalive requests every 90 seconds to prevent cold-start latency. Calibration runs at startup and every six hours, with non-blocking triggering when cold or warming states are encountered during quality gate evaluations.